

~~Experiments~~ Study of LAN Topology, Network devices and Address configuration.

Objective:-

- 1) To study the LAN topology used in College Campus.
- 2) To identify networking device & understand their function.
- 3) To analyze wired & wireless communication method used in the network.
- 4) To measure speed of Network for wired & wireless connections.
- 5) To obtain MAC, IP Address & Subnet mask.

→ Problem Statement - 1

- Draw & identify LAN topology of College Campus & clearly label all components such as router, switches & etc.

Solⁿ: The college campus network observed this study consists of:

- * Layer-3 Core switches in the server room.
- * Managed by POE+ switches.
- * Wireless Access point (APs)
- * Ethernet port in Classroom & Laboratories.

→ Network Topology

Hybrid Hierarchical Topology which combines:

- 1) Tree Topology.
- 2) Extended star Topology.
- 3) Ring Topology (server room).

→ The network hierarchical layer

- 1) Core layer.
- 2) Distribution layer.
- 3) Access layer.

* Core layer contains server room and server room have firewall, Router, 2-13 switches, Patch Rack, server, inverter, etc.

* Distribution layer contains 4-switches.

The 4 switches are →

- 1) Main Build → R7 (Room no-213, 2nd floor).
- 2) Main Build → R1 (Room no-301, 3rd floor).
- 3) Main Build → R6 (Room no-113, 1st floor).
- 4) Main Build → R2 (Room no-201, 2nd floor).

* This 4-switches contains →

i) 18 Port Gigabit Easy Smart switch with 16-Port. TL-SG1218MPE

ii) SG3428XMP → 24 Port Gigabit & 4-port 10GE SFP+ , L2+ Managed switches with 24 Port PoE+.

→ Now Let's discuss these 2 above components

1) TP-Link TL-SG1218MPE

* Likely used for: CCTV Camera, wifi access point, etc.

* 18 Port Gigabit "Easy Smart" switch.

* 16 Port support PoE+

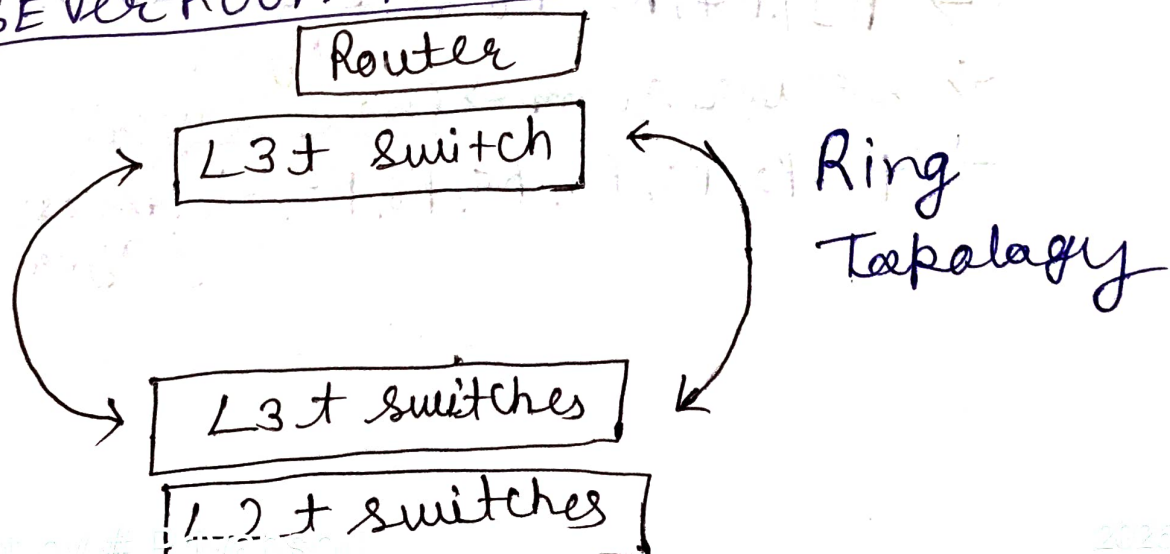
2) TP-Link Omada S/r3428xMP

- * Higher-end managed Layer 2+ switch.
- * 24 PoE+ Port, Uplinks, 10G SFP
- * Usually acts as a main distribution for each floor.

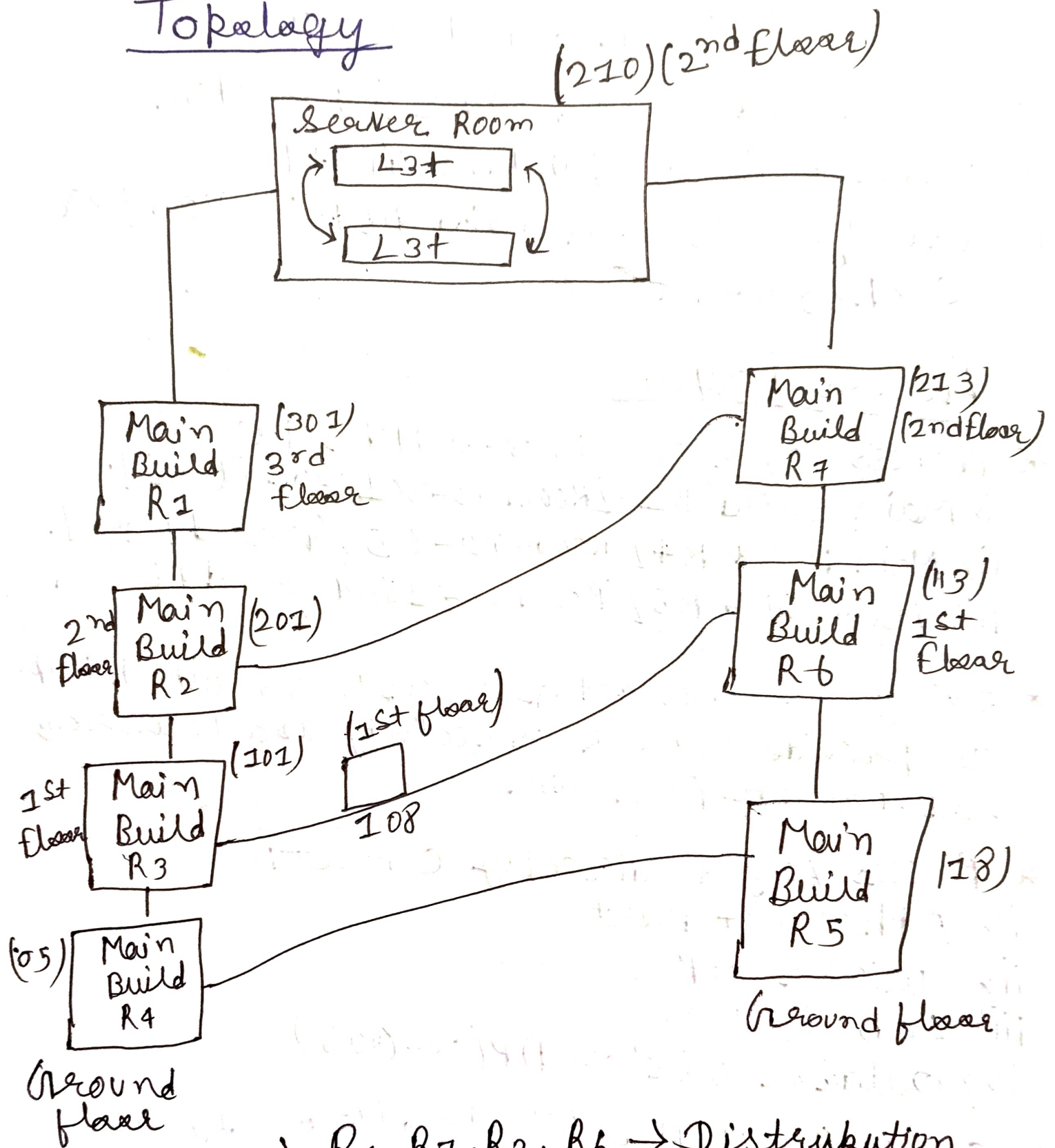
Access Layer

- * Access Layer contain 3-switches i.e.
 - i) Main Build R1 [Room no-301, 3rd floor].
 - ii) Main Build R4 [Room no-05, Ground floor].
 - iii) Main Build R5 [Room no-18, Ground floor]
- * Every Switches act as access Layer which helps to provide network connectivity to all classroom, Lab, Faculty Room, etc.
- * Access Layer directly connect:
 - i) Ethernet Port
 - ii) Wireless AP
 - iii) Lab System
 - iv) Outdoor wireless AP (ex-018)

→ SEVER ROOM Architecture/Topology

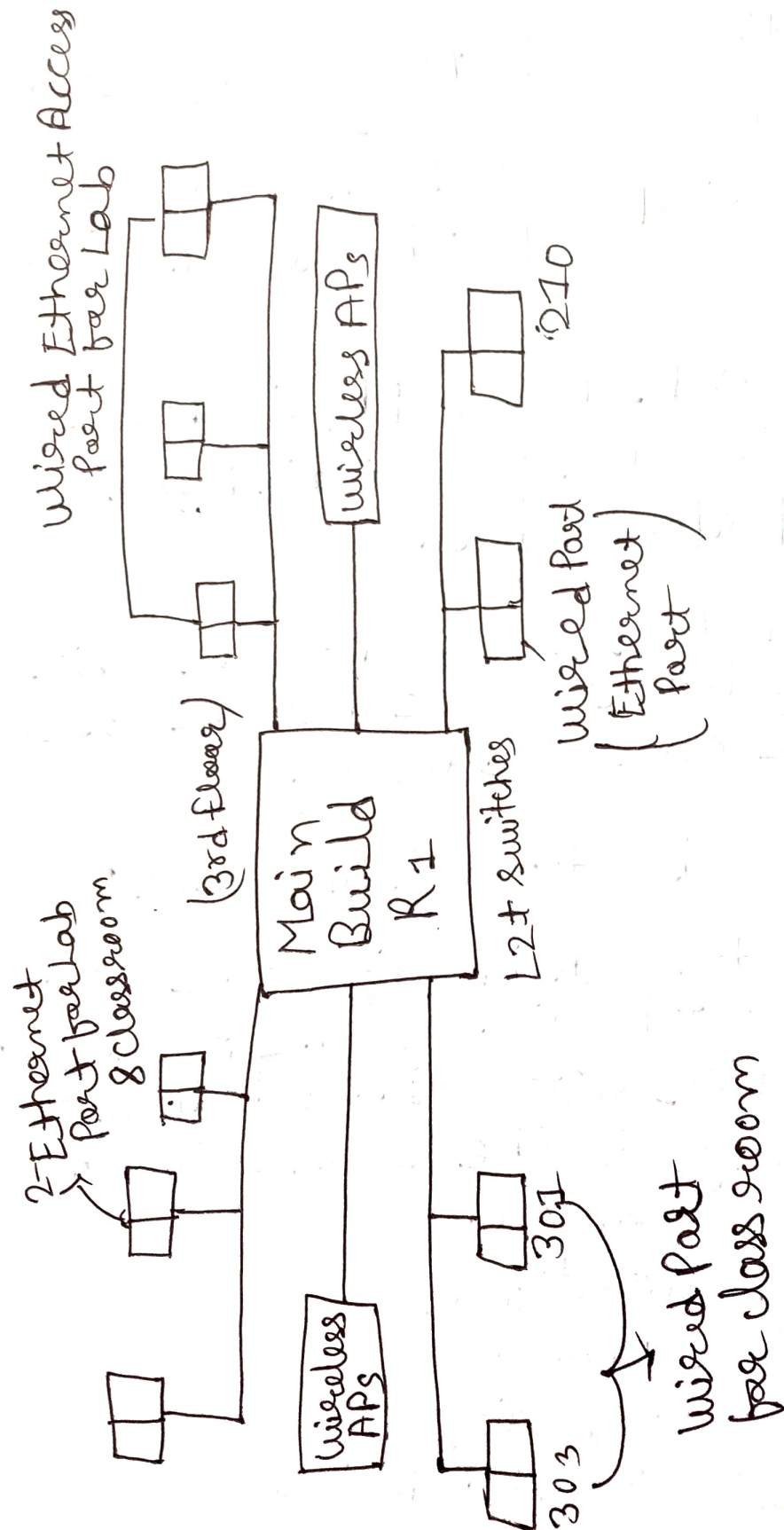


Topology



- R1, R7, R2, R6 → Distribution layer
- server room → Core layer
- R1, R2, R3, R4, R5, R6, R7, → Access Layer

Access Layer Switch (3rd floor)



Difference b/w wired & wireless speed?

Solⁿ:

Wired Network

- * faster Communication
- * Stable Connection
- * Low latency
- * Dedicated physical medium

→ Ethernet provides direct physical communication using network cables, resulting in higher speed & stable performance.

Wireless Network

- * slower communication
- * Affected by distance/Interference
- * Higher latency
- * Shared radio medium.

→ Wireless connections use radio waves which are affected by interference, distance, obstacles & multiple connected users.

8} Role of subnet mask in network?

Solⁿ: A subnet mask identifies the network portion & host portion of an IP address. It helps devices determine whether another device belongs to the same network or different network.

* It is important for:

- 1) Network segment
- 2) Efficient routing
- 3) Communication b/w devices

OBSERVATION

During the study of the college network infrastructure :

- i) A centralized server room was used as network back bone.
- ii) Two L3 switches were connected in redundant cyclic structure.
- iii) Managed TP-Link POE+ switches were used in network cabinets.

Advantage of Existing Network ?

- i) Scalable architecture.
- ii) Centralized management.
- iii) Supports POE+ device.
- iv) Easy expansion of floors & classroom.
- v) Support both wireless & wired.

Conclusion

The observed campus LAN follow a hybrid hierarchical topology consist of Tree, Extended star, Ring. The network uses managed POE switches, wireless access point, Ethernet port & centralized server infrastructure to provide efficient wired & wireless communication across the campus.

```
PS C:\Users\priya\Desktop> ipconfig
```

Windows IP Configuration

Ethernet adapter Ethernet 4:

```
Connection-specific DNS Suffix . :  
Link-local IPv6 Address . . . . . : fe80::b942:1116:4ec8:6dab%9  
IPv4 Address. . . . . : 192.168.56.1  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . :
```

Wireless LAN adapter Local Area Connection* 9:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter Local Area Connection* 10:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

Wireless LAN adapter WiFi:

```
Connection-specific DNS Suffix . : lan  
IPv6 Address. . . . . : 2409:40e4:108e:7902:fded:5b0f:d06d:a209  
Temporary IPv6 Address. . . . . : 2409:40e4:108e:7902:fd9d:2e72:697e:1c28  
Link-local IPv6 Address . . . . . : fe80::877c:7285:2f10:90a6%4  
IPv4 Address. . . . . : 192.168.31.225  
Subnet Mask . . . . . : 255.255.255.0  
Default Gateway . . . . . : fe80::feb0:deff:fe91:18c4%4  
                             192.168.31.1
```

Ethernet adapter Bluetooth Network Connection:

```
Media State . . . . . : Media disconnected  
Connection-specific DNS Suffix . :
```

```
PS C:\Users\priya\Desktop>
```


Performance

 Run new task ...



CPU
3% 2.00 GHz



Memory
7.6/15.3 GB (50%)



Disk 0 (C: D:)
SSD (NVMe)
9%



Ethernet
Ethernet 4
S: 0 R: 0 Kbps



WiFi
WiFi
S: 0 R: 0 Kbps

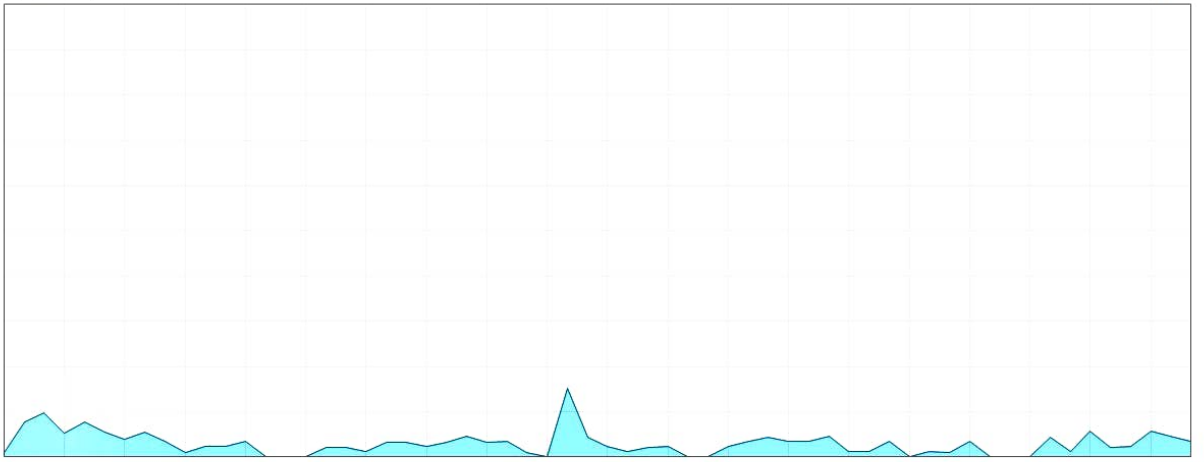


GPU 0
AMD Radeon (TM) Gr...
1% (44 °C)

CPU

AMD Ryzen 5 5625U with Radeon Graphics

% Utilisation 100%



60 seconds 0

Utilisation	Speed	Base speed:	2.30 GHz	
3%	2.00 GHz	Sockets:	1	
		Cores:	6	
Processes	Threads	Handles	Logical processors:	12
224	3009	108658	Virtualisation:	Enabled
Up time			L1 cache:	384 KB
3:05:45:58			L2 cache:	3.0 MB
			L3 cache:	16.0 MB